VNR VJIET

Name of Laboratory:

_____

Name of Experiment: _____

Experiment No: ____ Date: _____

Implement Data Link layer framing methods such as Character, Character stuffing and Bit stuffing

Bit Stuffing

Code:

```
// C program for Bit Stuffing (Data Link Layer)


#include <stdio.h>
#include <string.h>


// Function to perform bit stuffing
void bitStuffing(int N, int arr[])
{
    // Array to store stuffed bits
    int brr[30];

    // Index variables
    int i = 0; // for input array
    int j = 0; // for stuffed array
    int k;

    // Traverse the input array
    while (i < N)
    {
        // If current bit is 1
        if (arr[i] == 1)
        {
            int count = 1;
```

```
        // Store first 1
        brr[j] = arr[i];

        // Check next bits for consecutive 1s
        for (k = i + 1; k < N && arr[k] == 1 && count < 5; k++)
        {
            j++;
            brr[j] = arr[k];
            count++;

            // If 5 consecutive 1s found, insert 0
            if (count == 5)
            {
                j++;
                brr[j] = 0;
            }
        }

        // Move i to last checked position
        i = k;
    }
    else
    {
        // If bit is 0, directly copy
        brr[j] = arr[i];
        i++;
    }

    j++;
}
```

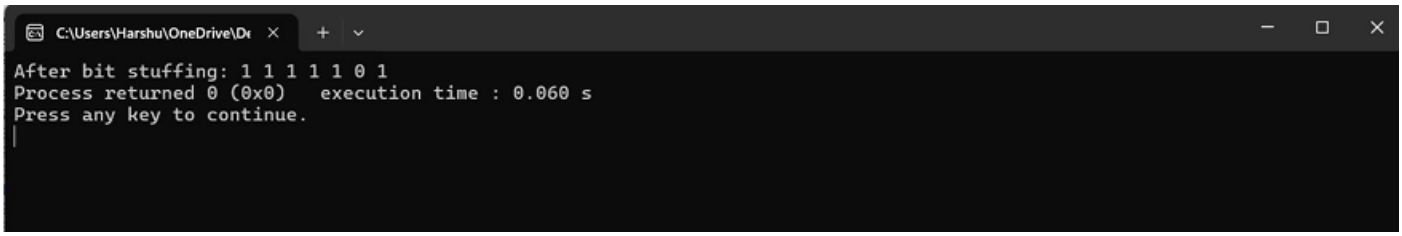
```
    // Print the stuffed bit sequence
    printf("Stuffed Bit Sequence: ");
    for (i = 0; i < j; i++)
    {
        printf("%d", brr[i]);
    }
}

// Driver Code
int main()
{
    int N = 6;
    int arr[] = {1, 1, 1, 1, 1, 1};

    bitStuffing(N, arr);

    return 0;
}
```

OUTPUT:



Character Stuffing

Code:

```
// C program for Character Stuffing and De-Stuffing


#include <stdio.h>
#include <string.h>


int main()
{
    char a[30];             // Input data
    char fs[50] = "";       // Stuffed frame
    char t[3];              // Temporary character
    char sd, ed;            // Starting and ending delimiters
    char x[3], s[3], d[3], y[3]; // Helper arrays
    int i, len;


    // Input original data
    printf("Enter characters to be stuffed: ");
    scanf("%s", a);


    // Input delimiters
    printf("\nEnter starting delimiter: ");
    scanf(" %c", &sd);


    printf("\nEnter ending delimiter: ");
    scanf(" %c", &ed);


    // Prepare delimiter strings
    x[0] = sd;   // start delimiter
    x[1] = '\0';
    y[0] = ed;   // end delimiter
```

```
y[1] = '\0';
// For stuffing if delimiter appears in data
s[0] = sd;
s[1] = sd;
s[2] = '\0';

d[0] = ed;
d[1] = ed;
d[2] = '\0';
// Add starting delimiter
strcat(fs, x);
// Perform stuffing
for (i = 0; i < strlen(a); i++)
{
   t[0] = a[i];
   t[1] = '\0';
   // If same as start delimiter → duplicate it
   if (a[i] == sd)
      strcat(fs, s);
   // If same as end delimiter → duplicate it
   else if (a[i] == ed)
      strcat(fs, d);
   // Otherwise copy as it is
   else
      strcat(fs, t);
}
```

```
 // Add ending delimiter
    strcat(fs, y);
    // Print stuffed frame
    printf("\nAfter Stuffing: %s", fs);
    // ------------------------------
    // Character De-Stuffing
    // ------------------------------
    len = strlen(fs);


// Remove starting delimiter
    for (i = 1; i < len - 1; i++)
    {
        fs[i - 1] = fs[i];
    }
    fs[i - 1] = '\0';
    // Print de-stuffed data
    printf("\nAfter De-Stuffing: %s", fs);
    return 0;
}
```

OUTPUT:

```
Enter characters to be stuffed:hellow

Enter a character that represents starting delimiter:C

Enter a character that represents ending delimiter:M

 After stuffing: ChellowM
 After De stuffing: hellow
```

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Implement on a Data set of characters the three CRC polynomials – CRC 12, CRC 16 and CRC CCIP

Code for generating CRC

```
// C program to implement CRC (Cyclic Redundancy Check)


#include <stdio.h>
#include <string.h>


#define N strlen(gen_poly)   // Length of generator polynomial


char data[28];              // Input data
char check_value[28];       // CRC value
char gen_poly[10];          // Generator polynomial


int data_length, i, j;


// Function to perform XOR operation
void XOR()
{
   // Perform XOR between check_value and generator polynomial
   for (j = 1; j < N; j++)
   {
      check_value[j] = (check_value[j] == gen_poly[j]) ? '0' : '1';
   }
}
// Function to calculate CRC
void crc()
```

```
{
   // Copy first N bits of data into check_value
   for (i = 0; i < N; i++)
   {
      check_value[i] = data[i];
   }

   do
   {
      // If leftmost bit is 1, perform XOR
      if (check_value[0] == '1')
      {
         XOR();
      }

      // Shift left by one bit
      for (j = 0; j < N - 1; j++)
      {
         check_value[j] = check_value[j + 1];
      }

      // Append next bit from data
      check_value[j] = data[i++];

   } while (i <= data_length + N - 1);
}

int main()
{
   printf("\nEnter data to be transmitted: ");
   scanf("%s", data);
```

```
printf("\nEnter the Generating Polynomial: ");
scanf("%s", gen_poly);

data_length = strlen(data);

// Append (N-1) zeros to data
for (i = data_length; i < data_length + N - 1; i++)
{
   data[i] = '0';
}
data[data_length + N - 1] = '\0';  // Proper string termination

printf("\n--------------------------------");
printf("\nData padded with n-1 zeros: %s", data);

// Calculate CRC
crc();

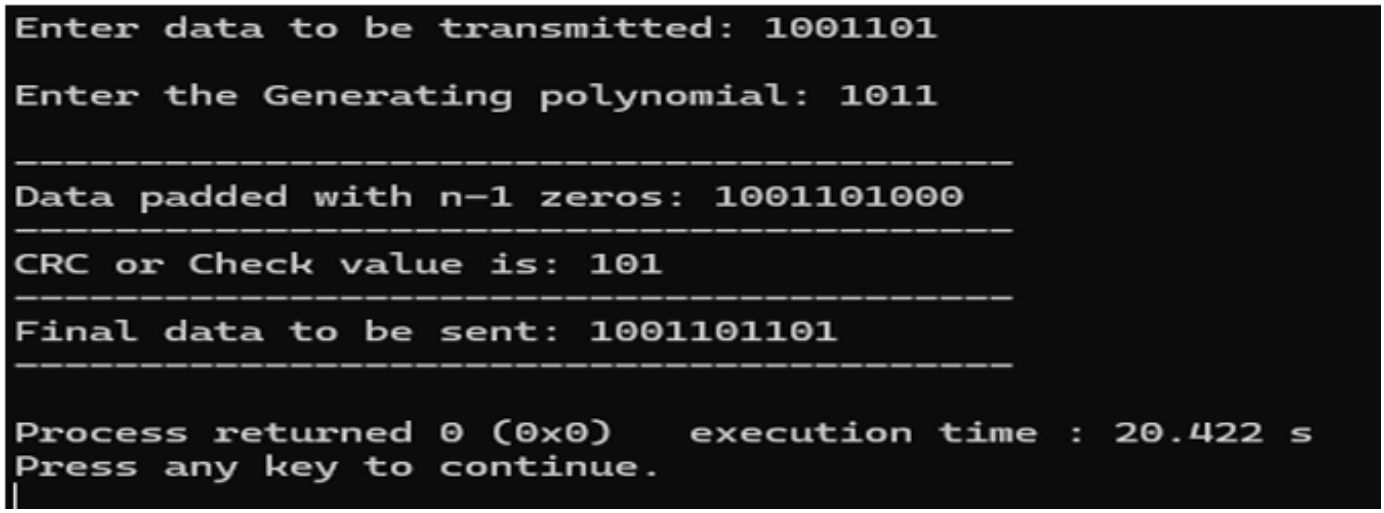
printf("\nCRC (Check Value): %s", check_value);

// Append CRC to original data
for (i = data_length; i < data_length + N - 1; i++)
{
   data[i] = check_value[i - data_length];
}
data[data_length + N - 1] = '\0';
```

```
    printf("\nFinal Data to be Sent: %s", data);
    printf("\n\n");

    return 0;
}
```

OUTPUT:



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CRC Error Detection Code

```
// C program for CRC (Sender + Receiver Side)


#include <stdio.h>
#include <string.h>


// Length of generator polynomial
#define N strlen(gen_poly)


// Data and CRC arrays
char data[28];          // Data to be transmitted / received
char check_value[28];   // CRC value
char gen_poly[10];      // Generator polynomial


// Variables
int data_length, i, j;


// Function to perform XOR operation
void XOR()
{
    // If bits are same → 0, else → 1
    for (j = 1; j < N; j++)
    {
        check_value[j] = (check_value[j] == gen_poly[j]) ? '0' : '1';
    }
}
```

```
// Function to perform CRC calculation
void crc()
{
   // Initialize check_value with first N bits
   for (i = 0; i < N; i++)
   {
      check_value[i] = data[i];
   }

   do
   {
      // If first bit is 1 → perform XOR
      if (check_value[0] == '1')
      {
         XOR();
      }

      // Left shift by 1 bit
      for (j = 0; j < N - 1; j++)
      {
         check_value[j] = check_value[j + 1];
      }

      // Append next bit from data
      check_value[j] = data[i++];

   } while (i <= data_length + N - 1);
}

// Receiver side: checks for error
void receiver()
```

```
{
   printf("\nEnter the received data: ");
   scanf("%s", data);

   printf("\nData received: %s\n", data);

   // Perform CRC again on received data
   crc();

   // Check if remainder is all zeros
   for (i = 0; (i < N - 1) && (check_value[i] != '1'); i++);

   if (i < N - 1)
      printf("\nError detected in received data!\n\n");
   else
      printf("\nNo error detected in received data.\n\n");
}

int main()
{
   // Input data
   printf("Enter data to be transmitted: ");
   scanf("%s", data);

   // Input generator polynomial
   printf("Enter the generating polynomial: ");
   scanf("%s", gen_poly);
```

```
    data_length = strlen(data);

    // Append (N-1) zeros to data
    for (i = data_length; i < data_length + N - 1; i++)
    {
        data[i] = '0';
    }
    data[data_length + N - 1] = '\0';

    printf("\nData after appending zeros: %s\n", data);

    // Calculate CRC
    crc();

    printf("CRC (Check Value): %s\n", check_value);

    // Append CRC to data
    for (i = data_length; i < data_length + N - 1; i++)
    {
        data[i] = check_value[i - data_length];
    }
    data[data_length + N - 1] = '\0';

    printf("Final data to be transmitted: %s\n", data);

    // Call receiver side
    receiver();

    return 0;
}
```

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OUTPUT:

```
Enter data to be transmitted: 1001101

 Enter the Generating polynomial: 1011
Enter the received data: 1001101101

-----------------------------
Data received: 1001101101
No error detected


Process returned 0 (0x0)   execution time : 26.581 s
Press any key to continue.
```

```
Enter data to be transmitted: 1001101

 Enter the Generating polynomial: 1011
Enter the received data: 1001001101

-----------------------------
Data received: 1001001101
Error detected


Process returned 0 (0x0)   execution time : 23.723 s
Press any key to continue.
```

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Basic Networking Commands

1. TRACERT:

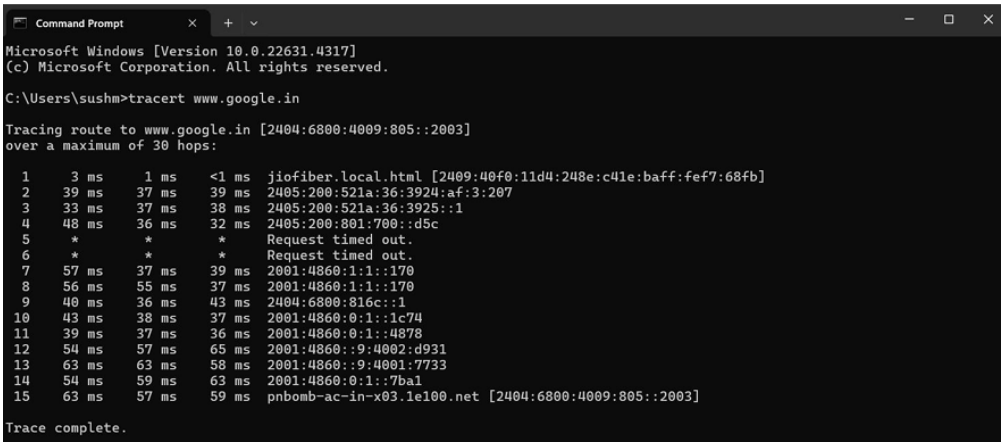
The tracert command prints the path. If all routers on the path are functional, this command prints the full path. If a router is down on the path, this command prints the path up to the last operational router.

This command is used to diagnose path-related problems. On an IP network, routers exchange IP packets between the source and the destination. They take IP packets from the source host and forward them in a sequence until they reach the destination host. The sequence of routers between the source and destination is known as the path. A path consists of all routers in a sequence that IP packets sent from the source host traverse to reach the destination host.

➢ Command:

tracert www.google.co.in

OUTPUT:



2. Ping

The ping command is used to test connectivity between two hosts. It sends ICMP echo request messages to the destination. The destination host replies with ICMP reply messages. If the ping command gets a reply from the destination host, it displays the reply along with round-trip times.

➢ Command:

ping google.com

OUTPUT:

```
C:\Users\sushm>ping google.com

Pinging google.com [2404:6800:4009:830::200e] with 32 bytes of data:
Reply from 2404:6800:4009:830::200e: time=62ms
Reply from 2404:6800:4009:830::200e: time=42ms
Reply from 2404:6800:4009:830::200e: time=68ms
Reply from 2404:6800:4009:830::200e: time=60ms

Ping statistics for 2404:6800:4009:830::200e:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 42ms, Maximum = 68ms, Average = 58ms
```

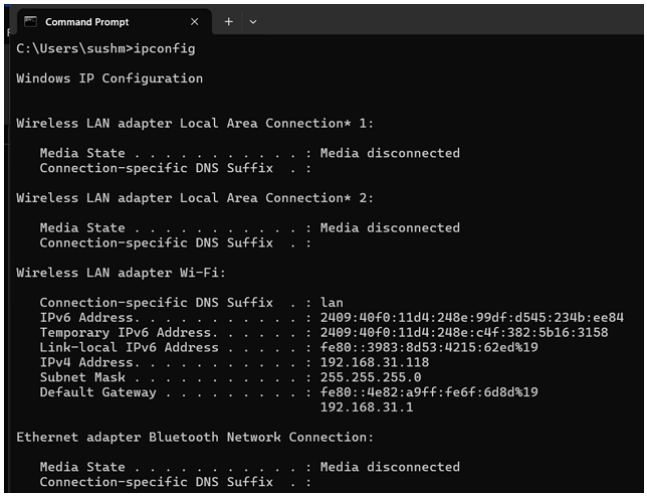
3. Ipconfig

This command displays all current TCP/IP network configuration values and refreshes Dynamic Host Configuration Protocol (DHCP) and Domain Name System (DNS) settings. This command is mainly used to view the IP addresses on the computers that are configured to obtain their IP address automatically.

➢ Command:

Ipconfig

OUTPUT:



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4. Arp

To send IP packets, a computer needs two addresses. These addresses are the MAC address and the IP address. A MAC address is the physical or hardware address of the NIC. An IP address is the logical or software address of NIC. If a computer knows the IP address of the destination computer but it does not know the MAC address of the destination computer, it uses the ARP protocol to know the MAC address of the destination computer.

The ARP protocol broadcasts a given IP address over a local network. The corresponding host responds to the broadcast with its MAC address. To avoid repetition, ARP stores the answer in a table known as ARP table. ARP maintains a separate ARP table for each NIC.

➢ Command:

arp

OUTPUT:

```
C:\Users\sushm>arp

Displays and modifies the IP-to-Physical address translation tables used by
address resolution protocol (ARP).

ARP -s inet_addr eth_addr [if_addr]
ARP -d inet_addr [if_addr]
ARP -a [inet_addr] [-N if_addr] [-v]

  -a            Displays current ARP entries by interrogating the current
                protocol data.  If inet_addr is specified, the IP and Physical
                addresses for only the specified computer are displayed.  If
                more than one network interface uses ARP, entries for each ARP
                table are displayed.
  -g            Same as -a.
  -v            Displays current ARP entries in verbose mode.  All invalid
                entries and entries on the loop-back interface will be shown.
  inet_addr     Specifies an internet address.
  -N if_addr    Displays the ARP entries for the network interface specified
                by if_addr.
  -d            Deletes the host specified by inet_addr. inet_addr may be
                wildcarded with * to delete all hosts.
  -s            Adds the host and associates the Internet address inet_addr
                with the Physical address eth_addr.  The Physical address is
                given as 6 hexadecimal bytes separated by hyphens. The entry
                is permanent.
  eth_addr      Specifies a physical address.
  if_addr       If present, this specifies the Internet address of the
                interface whose address translation table should be modified.
                If not present, the first applicable interface will be used.
Example:
  > arp -s 157.55.85.212   00-aa-00-62-c6-09  .... Adds a static entry.
  > arp -a                                    .... Displays the arp table.
```

5. Netstat

This command displays active connections, ports on which the computer is listening, Ethernet statistics, the IP routing table, and IP statistics.

The output of this command is organized in rows and columns. Each row represents a new connection or an entry in the output. It contains four columns. These columns provide the following information about the row.

➢ Command:

netstat

OUTPUT:

```
C:\Users\sushm>netstat

Active Connections

  Proto  Local Address          Foreign Address        State
  TCP    127.0.0.1:1521         sushmitha:49673        ESTABLISHED
  TCP    127.0.0.1:5939         sushmitha:61967        ESTABLISHED
  TCP    127.0.0.1:49673        sushmitha:1521         ESTABLISHED
  TCP    127.0.0.1:59432        sushmitha:59433        ESTABLISHED
  TCP    127.0.0.1:59433        sushmitha:59432        ESTABLISHED
  TCP    127.0.0.1:61967        sushmitha:5939         ESTABLISHED
  TCP    192.168.31.118:61971   au-mel-gcp-r002:5938   ESTABLISHED
  TCP    192.168.31.118:62097   20.249.115.164:https   ESTABLISHED
  TCP    192.168.31.118:62111   202:4070               ESTABLISHED
  TCP    192.168.31.118:62225   20.212.88.117:https    ESTABLISHED
  TCP    192.168.31.118:63056   52.182.143.208:https   TIME_WAIT
  TCP    192.168.31.118:63057   maa05s13-in-f3:https   ESTABLISHED
  TCP    192.168.31.118:63059   20.42.72.131:https     TIME_WAIT
  TCP    192.168.31.118:63068   whatsapp-cdn-shv-01-maa2:https  CLOSE_WAIT
  TCP    192.168.31.118:63073   13.89.179.11:https     TIME_WAIT
  TCP    192.168.31.118:63075   13.107.3.128:https     ESTABLISHED
  TCP    [2409:40f0:11d4:248e:c4f:382:5b16:3158]:58683  [2603:1040:a06:6::1]:https  ESTABLISHED
  TCP    [2409:40f0:11d4:248e:c4f:382:5b16:3158]:62054  [2603:1040:a06:6::1]:https  ESTABLISHED
  TCP    [2409:40f0:11d4:248e:c4f:382:5b16:3158]:62124  [2600:1901:1:304::]:https  ESTABLISHED
  TCP    [2409:40f0:11d4:248e:c4f:382:5b16:3158]:62160  [2600:1901:1:304::]:https  ESTABLISHED
  TCP    [2409:40f0:11d4:248e:c4f:382:5b16:3158]:62397  [2600:1901:1:7c5::]:https  ESTABLISHED
  TCP    [2409:40f0:11d4:248e:c4f:382:5b16:3158]:62638  sm-in-f188:5228        ESTABLISHED
  TCP    [2409:40f0:11d4:248e:c4f:382:5b16:3158]:62749  dns:https              ESTABLISHED
  TCP    [2409:40f0:11d4:248e:c4f:382:5b16:3158]:62757  maa03s31-in-x03:https  ESTABLISHED
  TCP    [2409:40f0:11d4:248e:c4f:382:5b16:3158]:62779  maa05s19-in-x0a:https  ESTABLISHED
  TCP    [2409:40f0:11d4:248e:c4f:382:5b16:3158]:62784  maa05s19-in-x0a:https  ESTABLISHED
  TCP    [2409:40f0:11d4:248e:c4f:382:5b16:3158]:62889  maa03s45-in-x02:https  ESTABLISHED
  TCP    [2409:40f0:11d4:248e:c4f:382:5b16:3158]:62895  maa03s40-in-x02:https  ESTABLISHED
  TCP    [2409:40f0:11d4:248e:c4f:382:5b16:3158]:62939  maa03s28-in-x03:https  ESTABLISHED
```

6. nslookup

The nslookup command queries the DNS to obtain domain names or IP address mapping. It's useful for verifying if a domain resolves correctly.

➢ Command:

nslookup www.google.com

OUTPUT:

```
C:\Users\sushm>nslookup www.google.com
Server:  jiofiber.local.html
Address:  192.168.31.1

Non-authoritative answer:
Name:    www.google.com
Addresses:  2404:6800:4009:803::2004
          142.250.183.36
```

6. dig

The dig (Domain Information Groper) command retrieves DNS records, such as A, MX, or TXT, for troubleshooting DNS issues.

➢ Command: dig www.google.com

7. curl

The curl command transfers data from or to a server using protocols like HTTP, FTP, or SCP. It is useful for testing API endpoints.

➢ Command: curl http://example.com

OUTPUT:



```
C:\Users\sushm>curl http://example.com
<!doctype html>
<html>
<head>
    <title>Example Domain</title>

    <meta charset="utf-8" />
    <meta http-equiv="Content-type" content="text/html; charset=utf-8" />
    <meta name="viewport" content="width=device-width, initial-scale=1" />
    <style type="text/css">
    body {
        background-color: #f0f0f2;
        margin: 0;
        padding: 0;
        font-family: -apple-system, system-ui, BlinkMacSystemFont, "Segoe UI", "Open Sans", "Helvetica Neue", Helvetica, Arial, sans-serif

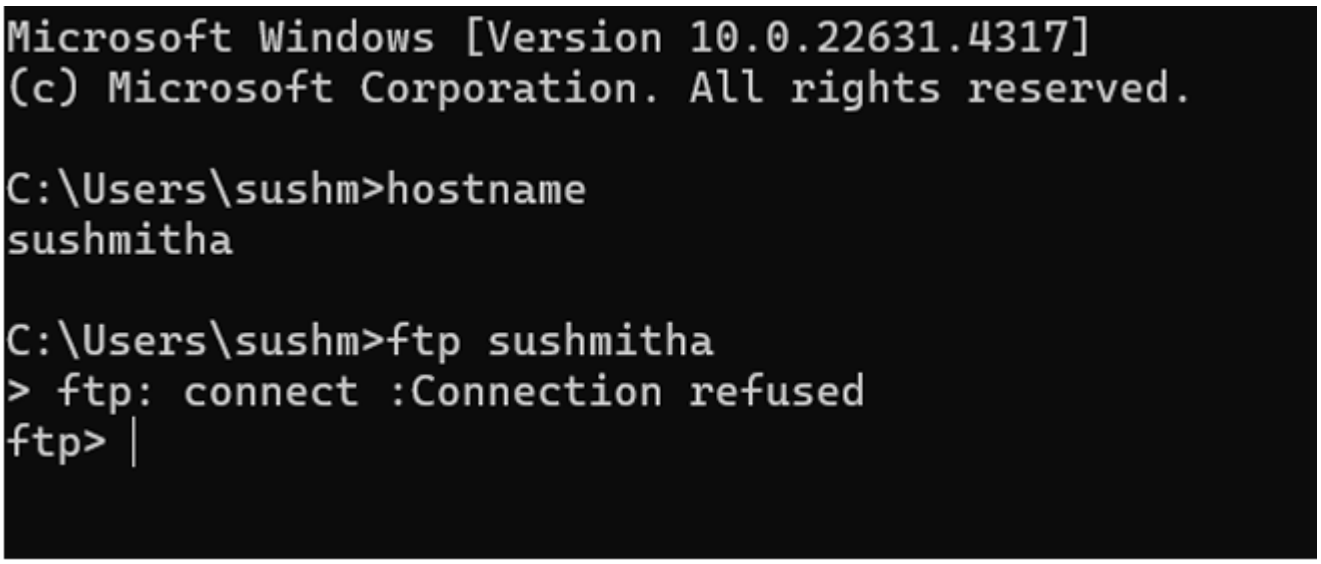
    }
    div {
        width: 600px;
        margin: 5em auto;
        padding: 2em;
        background-color: #fdfdff;
        border-radius: 0.5em;
        box-shadow: 2px 3px 7px 2px rgba(0,0,0,0.02);
    }
    a:link, a:visited {
        color: #38488f;
        text-decoration: none;
    }
    @media (max-width: 700px) {
        div {
            margin: 0 auto;
            width: auto;
        }
    }
    </style>
</head>
```

8. FTP (File Transfer Protocol)

Used to transfer files between computers on a network.

➢ Command: ftp <hostname>

OUTPUT:



```
Microsoft Windows [Version 10.0.22631.4317]
(c) Microsoft Corporation. All rights reserved.

C:\Users\sushm>hostname
sushmitha

C:\Users\sushm>ftp sushmitha
> ftp: connect :Connection refused
ftp>
```

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9. Netsh

A Windows utility for configuring and troubleshooting network settings.

➢ Command: netsh wlan show profiles

OUTPUT:

```
C:\Users\sushm>netsh wlan show profiles

Profiles on interface Wi-Fi:

Group policy profiles (read only)
---------------------------------
    <None>

User profiles
-------------
    All User Profile     : Jashwenth
    All User Profile     : Mannem_2G
    All User Profile     : AndroidAP
```

10. Route:

Displays or modifies the network routing table.

➢ Command: route-n

OUTPUT:

```
C:\Users\sushm>route -n

Manipulates network routing tables.

ROUTE [-f] [-p] [-4|-6] command [destination]
                  [MASK netmask]  [gateway] [METRIC metric]  [IF interface]

  -f           Clears the routing tables of all gateway entries.  If this is
               used in conjunction with one of the commands, the tables are
               cleared prior to running the command.

  -p           When used with the ADD command, makes a route persistent across
               boots of the system. By default, routes are not preserved
               when the system is restarted. Ignored for all other commands,
               which always affect the appropriate persistent routes.

  -4           Force using IPv4.

  -6           Force using IPv6.

  command      One of these:
                 PRINT     Prints  a route
                 ADD       Adds    a route
                 DELETE    Deletes a route
                 CHANGE    Modifies an existing route
  destination  Specifies the host.
  MASK         Specifies that the next parameter is the 'netmask' value.
  netmask      Specifies a subnet mask value for this route entry.
```

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Establishing a Network between computers :

Establishing a network between computers involves connecting multiple computers or devices so they can share resources like files, printers, or internet access. Below are steps to set up a basic network, whether for a small home network or a larger office setup:

I. Understand Your Networking Needs

➢ Type of Network: Determine if you need a wired (Ethernet) or wireless (Wi-Fi) network.

➢ Number of Devices: Assess how many devices will connect to the network and whether you need switches or additional access points.

II. Gather Necessary Equipment

➢ Router: For connecting multiple devices and sharing internet access.

➢ Switch: For expanding wired connections in larger networks.

➢ Ethernet Cables: For wired connections.

➢ Wireless Access Point (Optional): To extend the range of your wireless network.

➢ Computers and Devices: Ensure they have network adapters (wired or wireless).

III. Set Up a Wired Network

1. Connect the Router:

o Plug the router into a power source and connect it to your modem using an Ethernet cable.

2. Connect Computers to the Router:

o Use Ethernet cables to connect the computers to the router's LAN ports.

3. Test the Connection:

o Check each computer's network settings to ensure they obtain an IP address from the router (usually via DHCP).

4. Share Resources:

o Enable file and printer sharing on all computers.

IV. Set Up a Wireless Network

1. Configure the Router:

o Log in to the router's admin page (typically accessed via a web browser at an address like 192.168.1.1).

o Set a Wi-Fi name (SSID) and a strong password for security.

2. Connect Devices:

o On each computer, select the Wi-Fi network and enter the password.

3. Secure the Network:

o Use WPA3 or WPA2 encryption to protect your wireless network.

V. Assign Static or Dynamic Ips

➢ By default, most networks use DHCP to assign dynamic IP addresses.

➢ For better control in larger networks, you may assign static IPs to key devices like servers or printers.

VI. Test the Network

➢ Ping other computers on the network to check connectivity:

o ping <IP_address_of_other_computer>

➢ Use shared drives or printers to ensure everything works.

VII. Enable Resource Sharing

➢ On Windows:

o Go to Control Panel > Network and Sharing Center > Advanced Sharing Settings.

o Enable File and Printer Sharing and set network discovery to on.

➢ On macOS:

o Go to System Preferences > Sharing and select the resources to share.

➢ On Linux:

o Use tools like Samba for file sharing.

VIII. <u>Troubleshoot Common Issues</u>

➢ No Internet Access: Ensure the router is connected to the modem and the settings are correct.

➢ Device Not Detected: Check network adapter drivers or settings.

➢ Slow Speeds: Optimize router placement or upgrade to a gigabit router for wired connections.

VNR VJIET

Name of Laboratory:

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Configuring FTP server for file sharing

Configuring an FTP (File Transfer Protocol) server for file sharing involves setting up a server where files can

be uploaded, downloaded, and managed.

1. Understand FTP

• FTP Server: A computer that hosts files and allows users to upload or download them.

• FTP Clients: Applications used to connect to the server (e.g., FileZilla, WinSCP).

• Protocol: FTP can operate in plain text or securely using FTPS (SSL/TLS) or SFTP (SSH-based).

2. Prerequisites

• Server Machine: A dedicated computer or virtual machine to act as the FTP server.

• FTP Software: Examples include:

o Windows: IIS (Internet Information Services) or FileZilla Server.

o Linux: vsftpd, proftpd, or pure-ftpd.

• Network Configuration: Ensure the server has a static IP or domain name.

3. Install and Configure FTP Server

On Windows (Using FileZilla Server):

1. Download and Install FileZilla Server:

o Download FileZilla Server.

2. Configure Server:

o Launch FileZilla Server Interface.

o Set the host (IP address), port (default: 21), and admin password.

3. Create Users and Directories:

o Go to Edit > Users.

o Add a new user, set a password, and assign a directory.

4. Enable Firewall Rules:

o Open Control Panel > System and Security > Windows Defender Firewall.

o Add an inbound rule to allow traffic on port 21.

4. Secure the FTP Server

• Use FTPS or SFTP:

o Enable SSL/TLS for secure connections (FTPS).

o Alternatively, use SFTP if SSH is available.

• Limit Access:

o Restrict FTP users to specific directories.

o Disable anonymous access.

• Enable Strong Passwords:

o Require users to set strong passwords.

5. Connect to the FTP Server

• Using an FTP Client (e.g., FileZilla):

1. Open FileZilla Client.

2. Enter the server's IP address, username, password, and port (default: 21).

3. Connect to start transferring files.

6. Test the Server

• Test uploading and downloading files.

• Verify permissions to ensure users can only access their designated directories.

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Name of Laboratory:

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Implement Dijkstra's algorithm to compute the Shortest path through a graph.

Code:

```
// C program to implement Dijkstra's Algorithm


#include <stdio.h>


int main()
{
    int cost[10][10], i, j, n;
    int source, target;
    int visited[10] = {0};
    int dist[10], pre[10];
    int start, m, d, min;
    int path[10];


    // Input number of nodes
    printf("Enter number of nodes: ");
    scanf("%d", &n);


    // Input adjacency matrix
    printf("Enter weight of all the paths (Adjacency Matrix):\n");
    for (i = 1; i <= n; i++)
    {
        for (j = 1; j <= n; j++)
        {
            scanf("%d", &cost[i][j]);
```

```
        // If no direct edge, set cost to infinity (999)
        if (cost[i][j] == 0)
            cost[i][j] = 999;
    }
}

// Input source and target
printf("Enter the source node: ");
scanf("%d", &source);

printf("Enter the target node: ");
scanf("%d", &target);

// Initialize distances and predecessors
for (i = 1; i <= n; i++)
{
    dist[i] = 999;   // Initialize all distances to infinity
    pre[i] = -1;     // No predecessor initially
}

// Initialize source
visited[source] = 1;
dist[source] = 0;
start = source;

// Dijkstra's Algorithm
while (visited[target] == 0)
{
    min = 999;
    m = 0;
```

```
    for (i = 1; i <= n; i++)
    {
        // Calculate distance from current node
        d = dist[start] + cost[start][i];

        // Update distance if smaller and not visited
        if (d < dist[i] && visited[i] == 0)
        {
            dist[i] = d;
            pre[i] = start;
        }

        // Find next node with minimum distance
        if (min > dist[i] && visited[i] == 0)
        {
            min = dist[i];
            m = i;
        }
    }

    start = m;
    visited[m] = 1;
}

// Path reconstruction
start = target;
```

```
    j = 0;

    while (start != -1)
    {
        path[j++] = start;
        start = pre[start];
    }

    // Print shortest path
    printf("Shortest Path: ");
    for (i = j - 1; i >= 0; i--)
    {
        if (i != j - 1)
            printf(" -> ");
        printf("%d", path[i]);
    }

    // Print shortest distance
    printf("\nShortest distance = %d\n", dist[target]);

    return 0;
}
```

OUTPUT:

```
Enter number of nodes
 4
Enter weight of all the paths in adjacency matrix form
0 10 30 100
10 0 10 90
30 10 0 30
100 90 30 0
Enter the source
1
Enter the target
4
1 -> 2 -> 3 -> 4
 shortest path is 50
Process returned 0 (0x0)   execution time : 55.257 s
Press any key to continue.
```

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Network Topologies

Star Topology

To perform the simulation, follow the below given steps:

1ct the topology from the dropdown that you would like to implement.

2. Click on either PC or laptop (present at the bottom) to place the device on the working area.

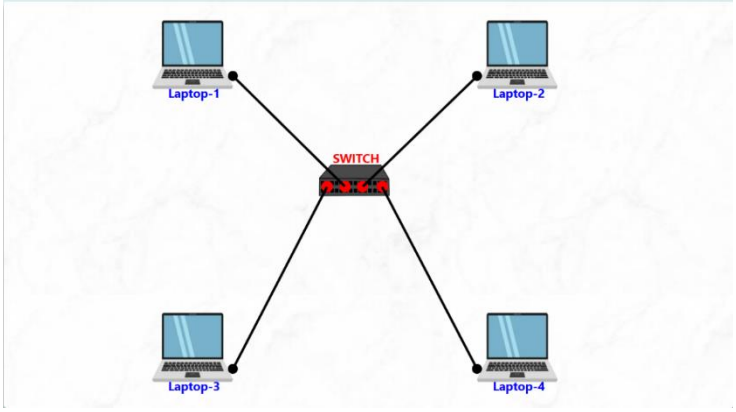
- You can create a topology using any number of end devices, but here we will use a total of four.

3. Now click on either switch or hub.

4. Connect all the end devices with the central device as shown in Fig. 1 (click on the image for a better view).

NOTE: Click a device port and drag and drop it to another device port to connect.

5. Once the topology is constructed, click on the check button to verify if the constructed topology is built correctly or not.



Bus Topology

To perform the simulation, follow the below given steps:

1. Click on either PC or laptop (present at the bottom) to place them on the working area.

- You can create topology using any number of end devices, but here we will use a total of four.

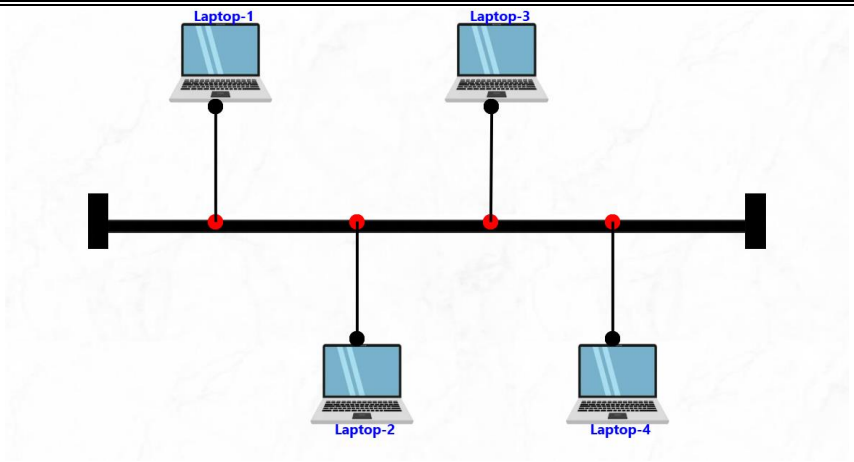
2. Click on cable.

3. Click twice on the terminator.

4. Now connect end devices with the main cable as shown in Fig. 1 (click on the image for a better view).

NOTE: Click device port and drag and drop it to the port of the cable you want to connect.

5. Once the topology is constructed, click on the check button to verify if the constructed topology is built correctly or not.



Ring Topology

To perform the simulation, follow the below given steps:

1. Click on either PC or laptop (present at the bottom) to place the device on the working area.

2. Now click on either switch or hub.

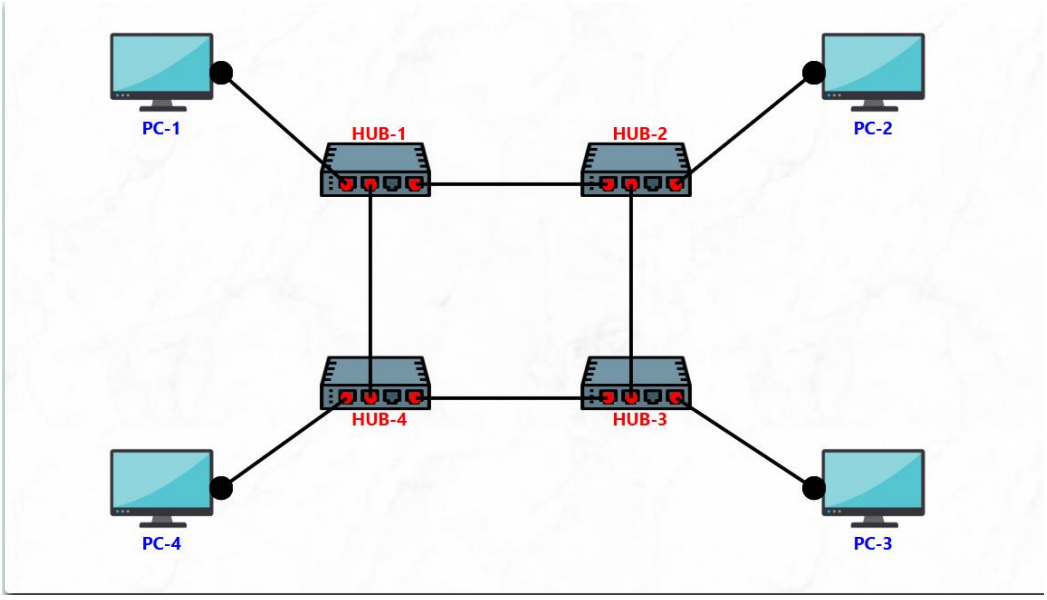
3. Now create a communication link as shown in Fig. 1 (click on the image for a better view).

3.1. Connect each pair of end devices with its corresponding intermediate device, e.g., PC1 with Hub1, PC2 with Hub2, and so on.

3.2. Next, connect switches/hubs together, e.g., Hub1 with Hub2, Hub2 with Hub3, and so on.

NOTE: Click a device port and drag and drop it to another device port to connect.

4. Once the topology is constructed, click on the check button to verify if the constructed topology is built correctly or not.



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Mesh Topology

To perform the simulation, follow the below given steps:

1. Click on the PC/laptop (present at the bottom) to place the device on the working area.

2. Next click on the switch or hub.

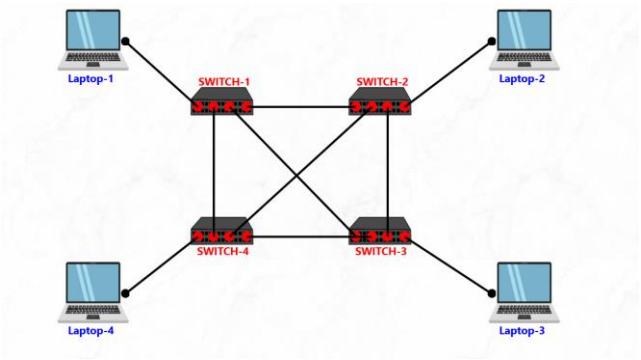
3. Now create a communication link as shown in Fig. 1 (click on the image for a better view).

3.1. Connect each of the end devices with the corresponding switch/hub turn by turn, e.g., PC1 with Hub1, PC2 with Hub2, and so on.

3.2. Now connect each switch/hub with the remaining three switches/hubs in your mesh network.

NOTE: Click a device port and drag and drop it to another device port to connect.

4. Once the topology is constructed, click on the check button to verify if the constructed topology is built correctly or not.



Hybrid Topology

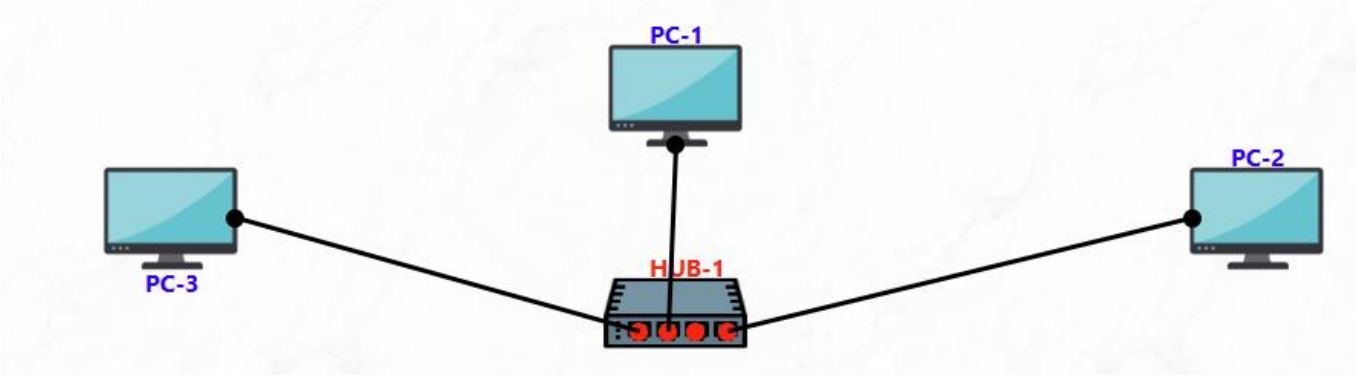


Fig. A: Star topology

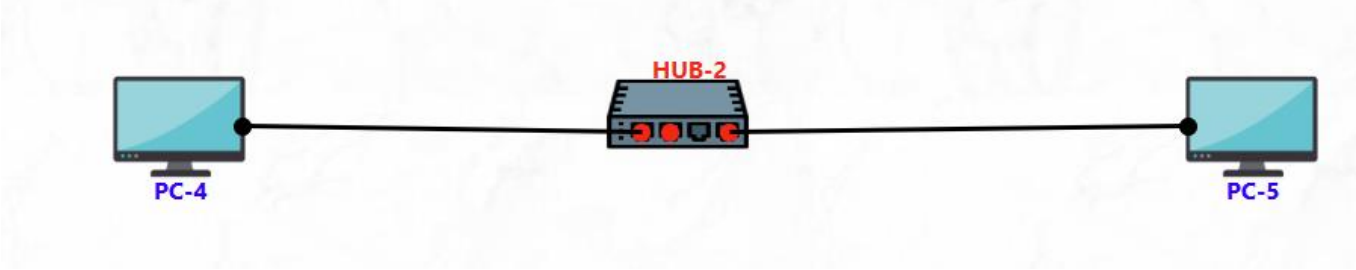


Fig. B: Bus topology

We are building a **hybrid** of these two topologies: **STAR + BUS**.

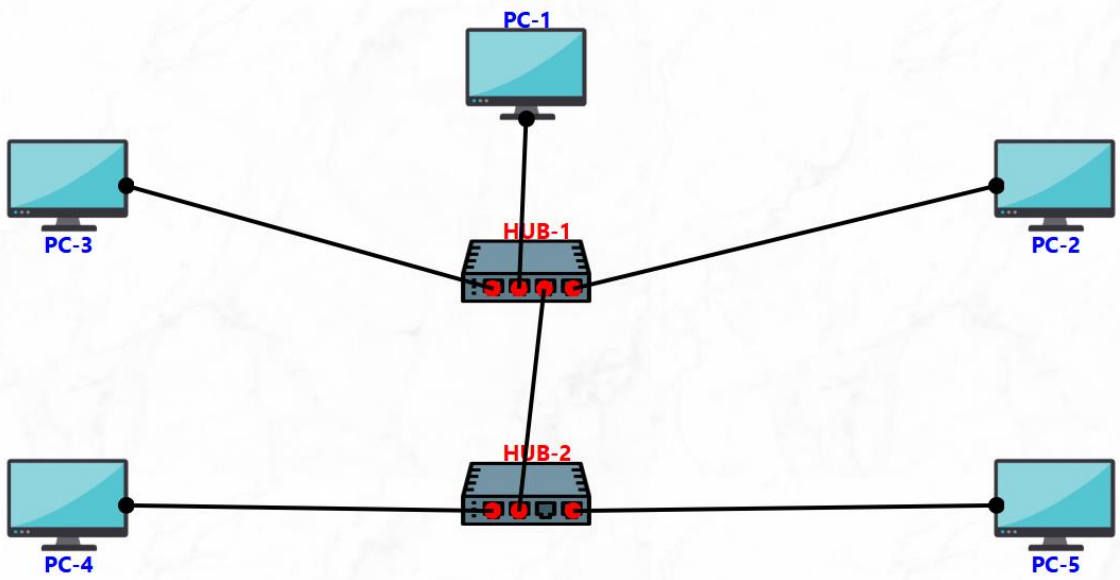
Steps:

1. **To perform the simulation (choose one from each category):**
   End devices: *PC* or *Laptop*
   Central devices: *Hub* or *Switch*

2. Wire to match the diagram: (1,2,3) → *device-1*, *device-1* ↔ *device-2*, (4,5) → *device-2*.

Note: Click a device port and drag to another device port to connect

3. Click *Check* to validate the hybrid topology.



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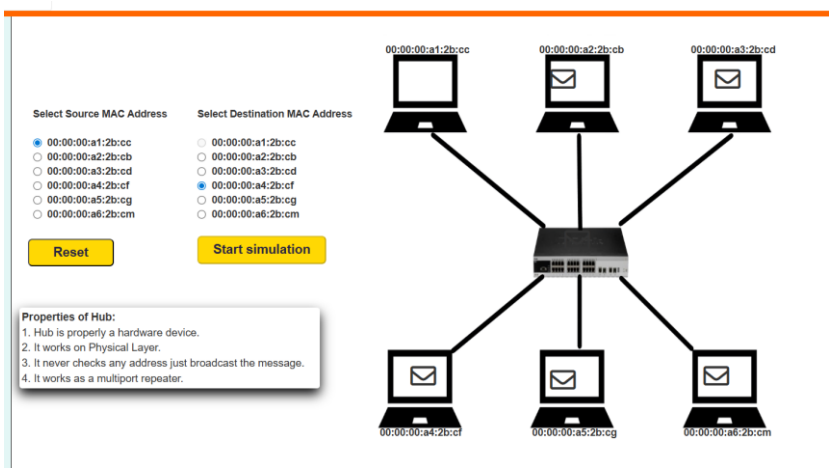
Experiment No: ____ Date: _____

Demonstration of Network Devices

Hub demonstration:

Follow these steps:

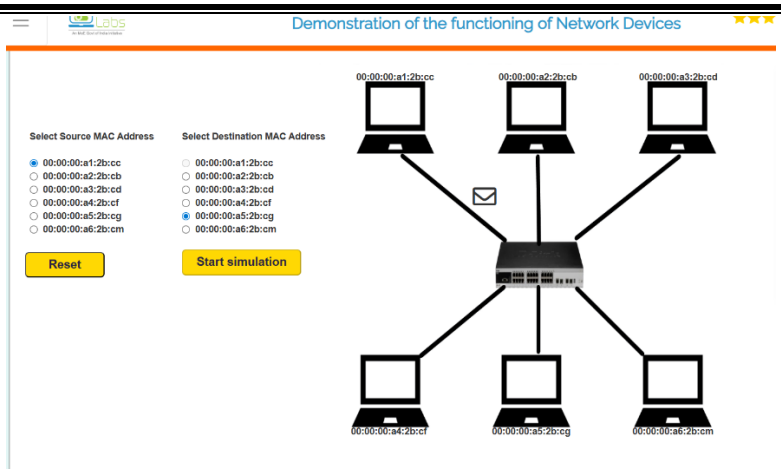
1. In the Simulation Frame, select the Source MAC Address and the Destinaton MAC Address.
2. To begin the simulation, click the **'Start Simulation'** button.
3. To simulate again, first click the **'Reset'** button.
4. Try out the other devices just by clicking on the dropdown visible at the top stating Choose a Network Device.
5. If you want to exit the simulation just close the tab.

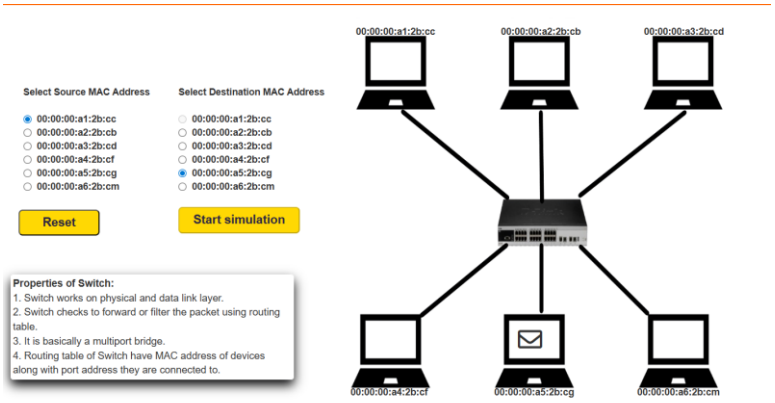


Switch:

Follow these steps:

1. In the Simulation Frame, select the Source MAC Address and the Destinaton MAC Address.
2. To begin the simulation, click the **'Start Simulation'** button.
3. To simulate again, first click the **'Reset'** button.
4. Try out the other devices just by clicking on the dropdown visible at the top stating Choose a Network Device.
5. If you want to exit the simulation just close the tab.





Router:

Follow these steps:

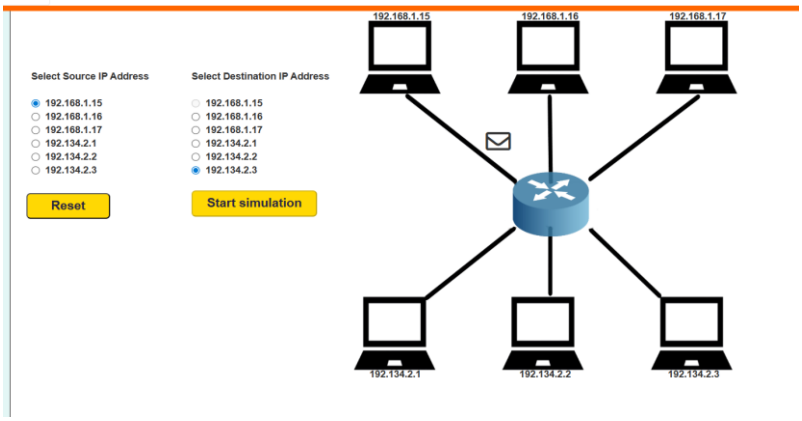
1. In the Simulation Frame, select the Source IP Address and the Destinaton IP Address.

2. To begin the simulation, click the **'Start Simulation'** button.

3. To simulate again, first click the **'Reset'** button.

4. Try out the other devices just by clicking on the dropdown visible at the top stating Choose a Network Device.

5. If you want to exit the simulation just close the tab.



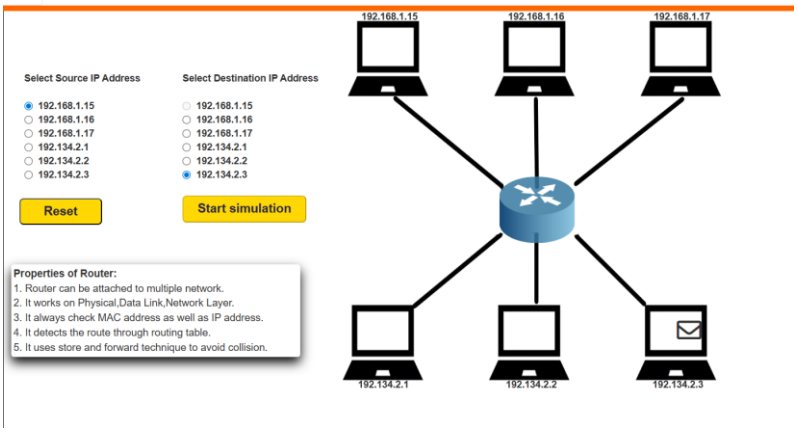
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Bridge:

Follow these steps:s

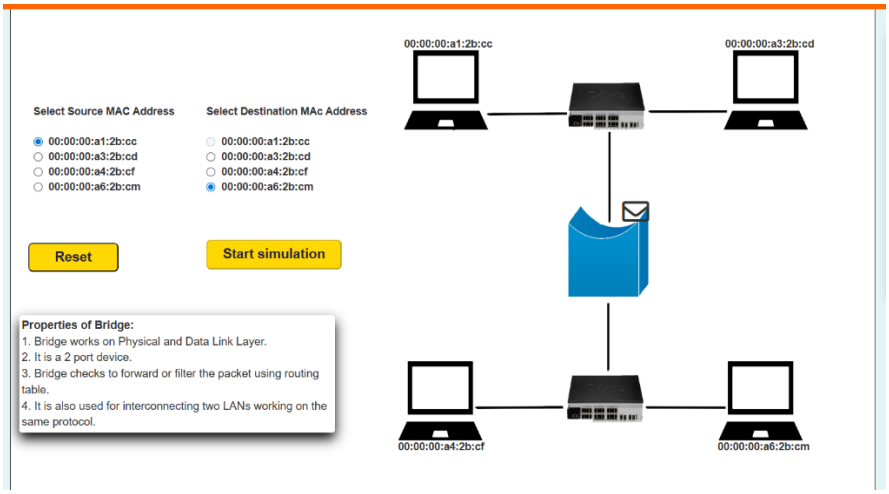
1. In the Simulation Frame, select the Source MAC Address and the Destinaton MAC Address.

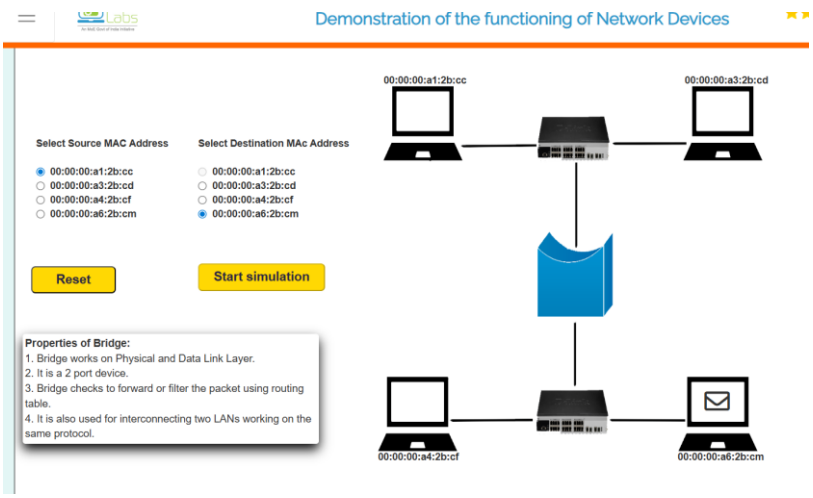
2. To begin the simulation, click the **'Start Simulation'** button.

3. To simulate again, first click the **'Reset'** button.

4. Try out the other devices just by clicking on the dropdown visible at the top stating Choose a Network Device.

5. If you want to exit the simulation just close the tab.




Demonstration of the functioning of Network Devices
Select Source MAC Address
00:00:00:a1:2b:cc
00:00:00:a3:2b:cd
00:00:00:a4:2b:cf
00:00:00:a6:2b:cm
Select Destination MAc Address
00:00:00:a1:2b:cc
00:00:00:a3:2b:cd
00:00:00:a4:2b:cf
00:00:00:a6:2b:cm
Reset
Start simulation
Properties of Bridge:
1. Bridge works on Physical and Data Link Layer.
2. It is a 2 port device.
3. Bridge checks to forward or filter the packet using routing table.
4. It is also used for interconnecting two LANs working on the same protocol.
00:00:00:a1:2b:cc
00:00:00:a3:2b:cd
00:00:00:a4:2b:cf
00:00:00:a6:2b:cm